

# Computer Networks Viva Question Sets

## Set 1:

1. Explain how CIDR reduces routing table size.
2. Design a VLSM scheme for subnets needing 100, 50, 25, and 10 hosts.
3. What is the difference between distance vector and link state routing?
4. How does a standard ACL decide which packets to filter?
5. Can a router connect two VLANs? How?

## Set 2:

1. Subnet 192.168.10.0/24 into 4 equal parts using CIDR. What will be the new mask?
2. Why is static routing not suitable for large networks?
3. How does RIP handle routing loops?
4. Write a command to deny a subnet using a standard ACL.
5. What is the role of a trunk port in VLAN?

## Set 3:

1. Define supernetting. When is it used?
2. Plan a VLSM for a network with departments needing 120, 60, 30, and 5 IPs.
3. What happens if two routes with the same prefix exist, one static and one RIP?
4. How does an extended ACL differentiate traffic?
5. What are native VLANs, and what are their security implications?

## Set 4:

1. What is the major advantage of CIDR over classful addressing?
2. Convert 255.255.255.224 to prefix notation and find total hosts.
3. How does EIGRP calculate metric?
4. What is the wildcard mask used for 192.168.1.0/24?
5. Explain the process of inter-VLAN routing using sub-interfaces.

## Set 5:

1. Given a CIDR block of /22, how many IP addresses are usable?
2. Why is VLSM better than fixed-length subnetting?
3. Differentiate between RIP and OSPF.

4. What happens when you place an ACL inbound on an interface?
5. What is the purpose of VLAN tagging (802.1Q)?

#### **Set 6:**

1. How does CIDR affect route summarization in BGP?
2. What problems can occur if VLSM is not planned properly?
3. Why is RIP considered slow in convergence?
4. How does a router match ACL rules (first match vs last)?
5. What is the difference between access port and trunk port?

#### **Set 7:**

1. Design a subnet plan for 10.0.0.0/8 using CIDR for 8 departments.
2. When should you use static over dynamic routing?
3. How does OSPF elect a DR and BDR?
4. Write a basic extended ACL to deny HTTP traffic.
5. How can you test if inter-VLAN routing is working?

#### **Set 8:**

1. Convert /28 to dotted decimal and find number of subnets in a /24.
2. How do you prevent routing loops in RIP?
3. How does EIGRP differ from RIP in terms of update transmission?
4. Give a real-life example of when to use an extended ACL.
5. Can a VLAN span multiple switches?

#### **Set 9:**

1. Can two networks with CIDR blocks /25 and /26 overlap?
2. In VLSM, why is it better to allocate larger subnets first?
3. Compare administrative distance of static, RIP, and EIGRP.
4. Where should standard ACLs be placed in the network?
5. Explain how a switch handles tagged and untagged VLAN traffic.

#### **Set 10:**

1. What is the subnet mask for /23 and how many usable hosts does it provide?
2. How does VLSM support efficient IP usage?
3. What is the difference between IGP and EGP? Name examples.

4. Write a command to allow only a specific host using a standard ACL.
5. What is inter-switch VLAN communication and how is it enabled?

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2. Plan a VLSM for different department sizes. (Set 11)
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